

Test III

- I) Write your name, the date, and the course title (i.e. Precalculus).
- II) Show all work.
- III) Draw a rectangle around your final answer for each problem.
- IV) A graphing calculator is required, but is not allowed for some problems.

① Each table represents a relation that has the indicated symmetry. Fill in the missing parts of the table.

i) Symmetric about the X-axis.

X	2	-2	3	-3	4														
Y	-2	2	-3	3	-4														

ii) Symmetric about the Y-axis.

X	9	11	-2	-1	14														
Y	-9	-11	0	3	-14														

② i) Convert the equation below to rectangular form.

$$\sin \theta = r$$

ii) Determine if the function below is even, odd, or neither.

$$f(x) = 3^x + 4$$

③ i) Create parametric equations for the relation below. Use "t" as the parameter.

$$y - 1 = 4^x$$

ii) Determine if the function below is even, odd, or neither.

$$f(x) = \frac{1}{x^2} + 10x^4$$

④ The equation of a relation is written below. Write the equation of each reflection. Simplify if possible.

$$\frac{\csc x}{x} + \frac{y}{\cot y} = 1$$

i) Reflect across the origin →

ii) Reflect across the y-axis →

Is the relation symmetric about the

iii) origin?

iv) y-axis?

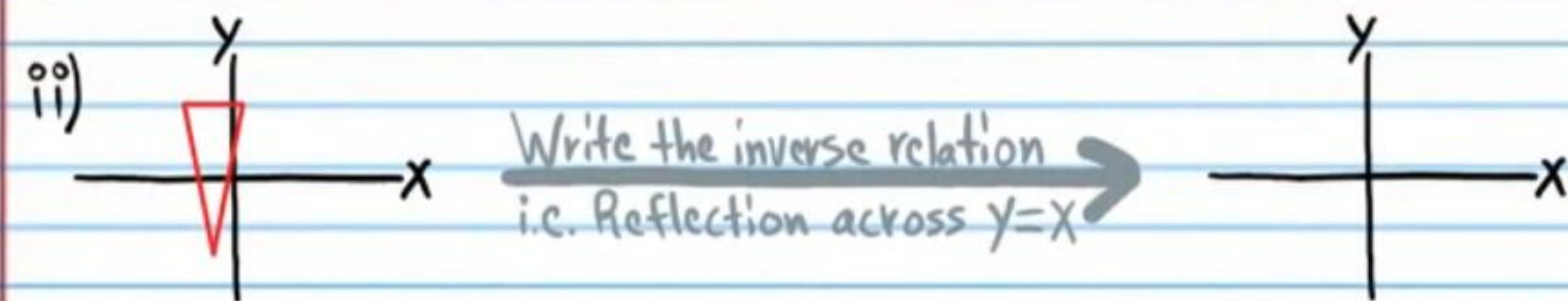
⑤ Relations are written below in various forms. Write each inverse relation in its corresponding form.

i)

x	100	22	-7	-11	20
y	40	33	4	-11	21

 Write the inverse relation
i.e. Reflection across $y=x$

x					
y					



iii) $2^y = x + 18$ Write the inverse relation
i.e. Reflection across $y=x$

iv) $|x-4| = |y-1|$ Write the inverse relation
i.e. Reflection across $y=x$

⑥ Solve.

i) $1 = \sqrt[3]{4^x} \cdot 16^x$

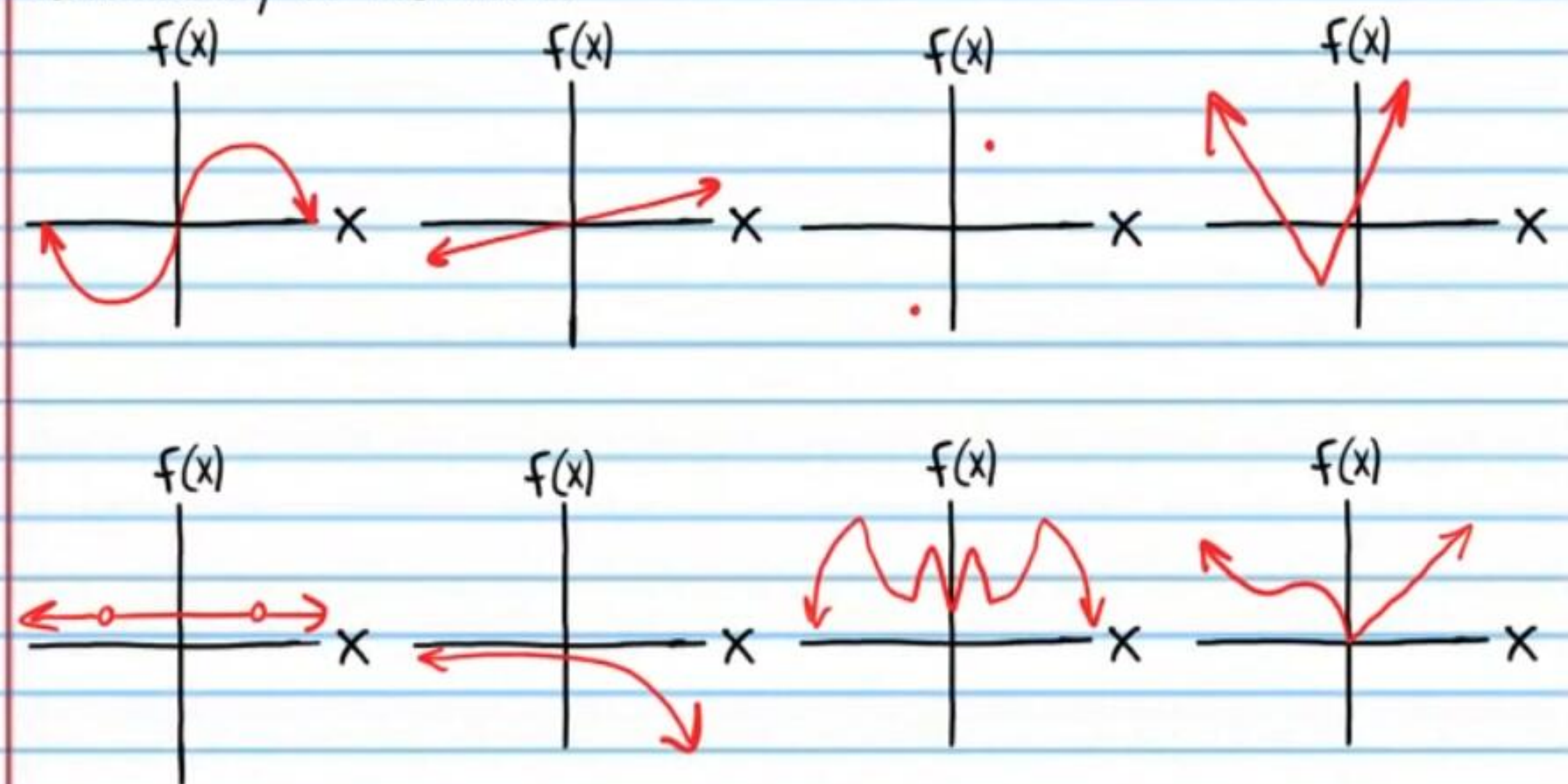
ii) $4 = 2 + 2 \log_x 7$

⑦ i) "f" and "g" are functions represented by the tables below. If "f" is an even function and "h" is an odd function, fill the empty spaces with other values of the functions.

x	6	10	-10	-16	-2					
f(x)	-1	2	0	4	-3					

x	15	9	-3	-8	-7					
h(x)	14	5	-3	-8	-1					

ii) Determine if the graphs below appear to be even functions, odd functions, or neither.



⑧ A relation is written below in table form. Perform the following reflections and fill in the tables.

i)

x	23	44	-6	-20	-3
y	34	20	-2	-30	19

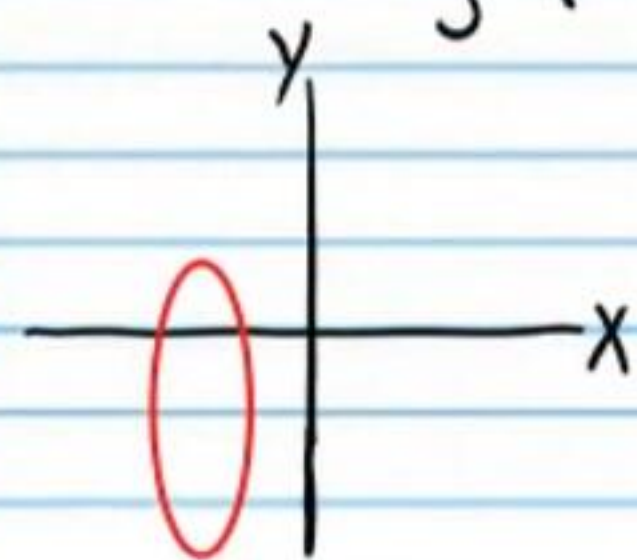
 Reflect across the x-axis

x					
y					

ii) Reflect across the y-axis

x					
y					

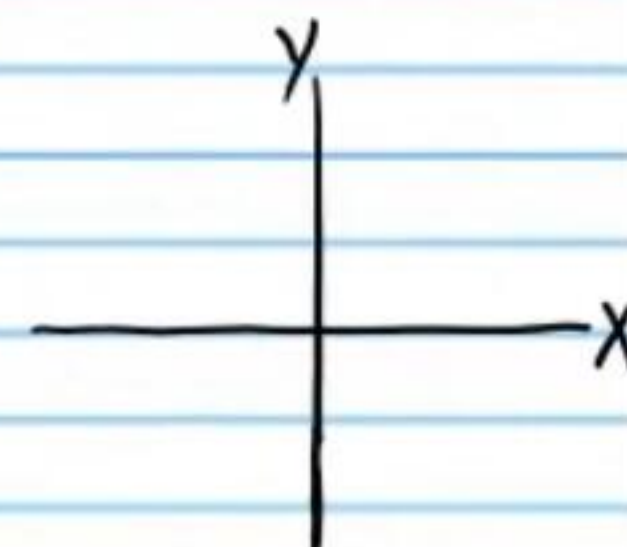
A relation is written below in the form of a graph.
Draw the graphs of the indicated reflections.



iii) Reflect across
the x-axis →



iv) Reflect across
the origin →



⑨ Solve. * You must use your calculator. Round to the nearest hundredth.

$$3(5^x) - 2 = 4$$

Convert the following polar coordinates to rectangular coordinates.
Round to the nearest thousandth if necessary.

↙ You may not use a calculator. ↘

⑩ $(8\sqrt{2}, 315^\circ)$

⑪ $(5, -90^\circ)$

↪ You may not use a calculator.

⑫ $(7, -300^\circ)$

⑬ $(-10, 203^\circ)$

Convert the rectangular coordinates below to polar coordinates.
Use degree units for angle measures. Round to the nearest thousandth if necessary.

⑭ $(-1, -2)$

⑮ $(3, -2)$

You may not use a calculator ↴

⑩ $(-9\sqrt{3}, 9)$

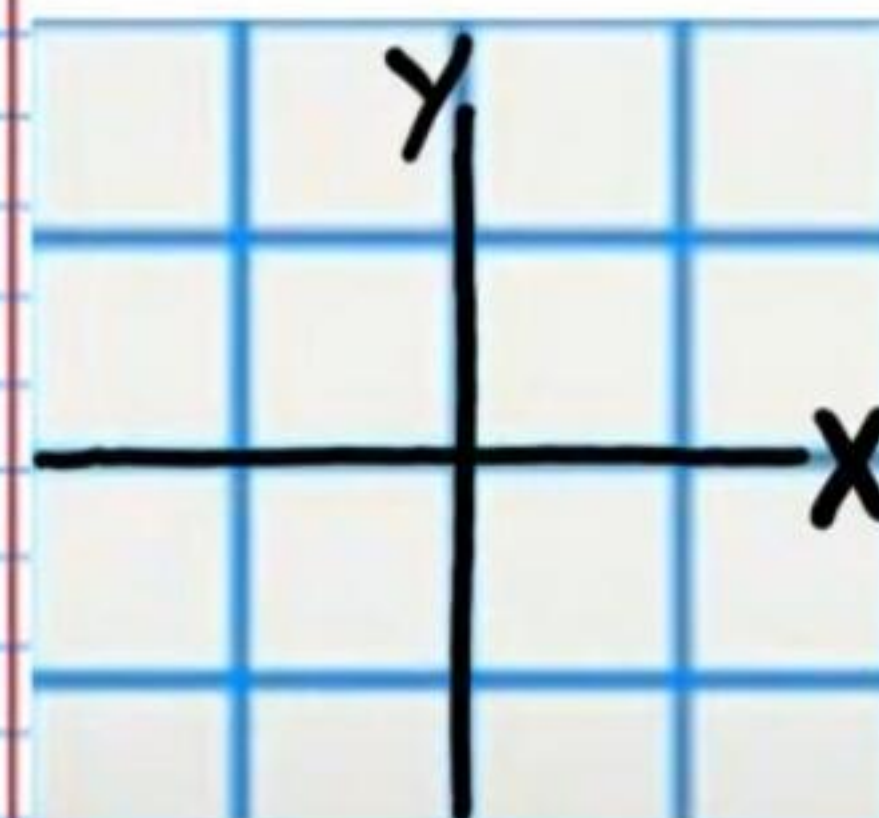
⑪ $(11, 0)$

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot.

⑫ $r = 2\sin(2\theta)$

1st Point

2nd Point



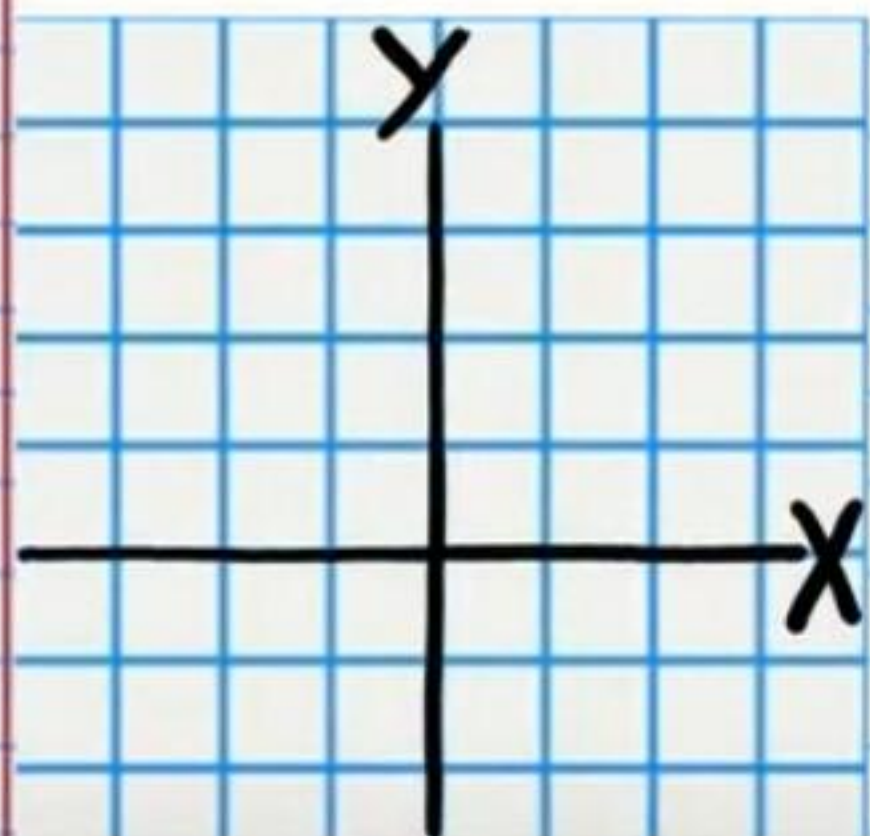
3rd Point

4th Point

①⑨ $r = \frac{2}{1+2\sin\theta}$

1st Point

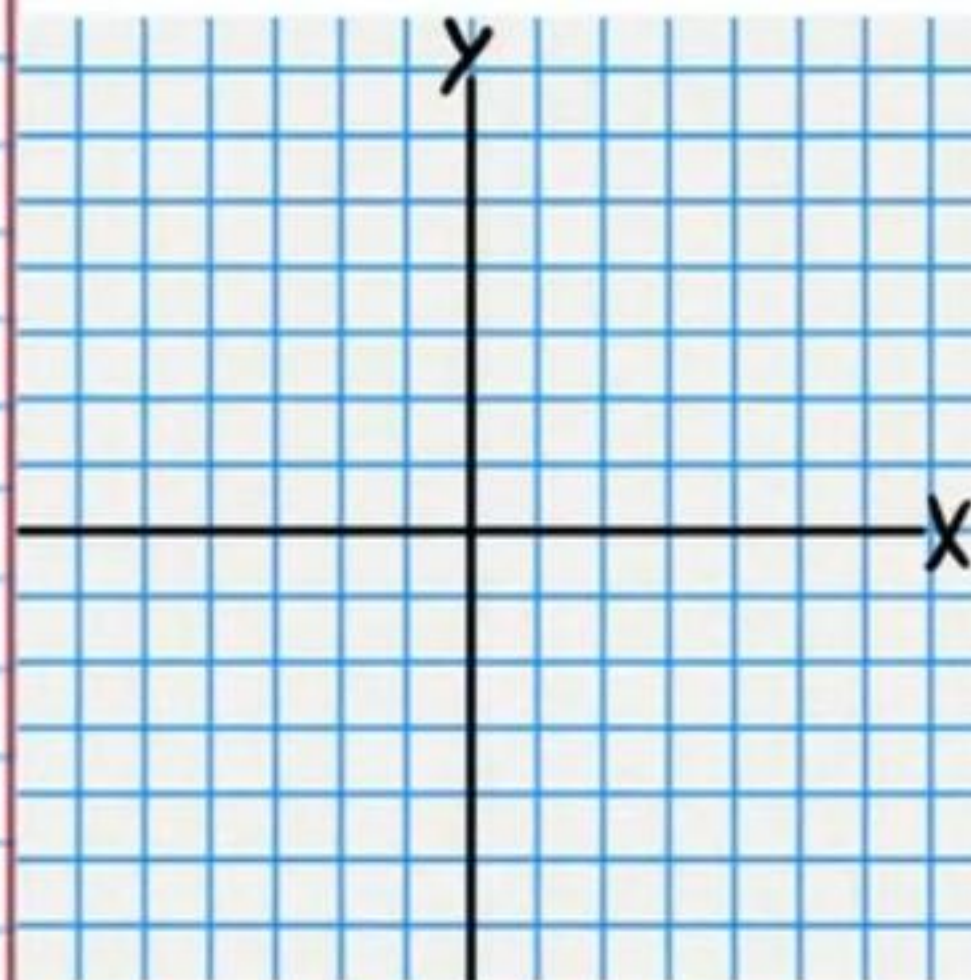
2nd Point



3rd Point

4th Point

②⑩ $r = 6$



②① Convert the equation below to polar form. *Simplify if possible.*

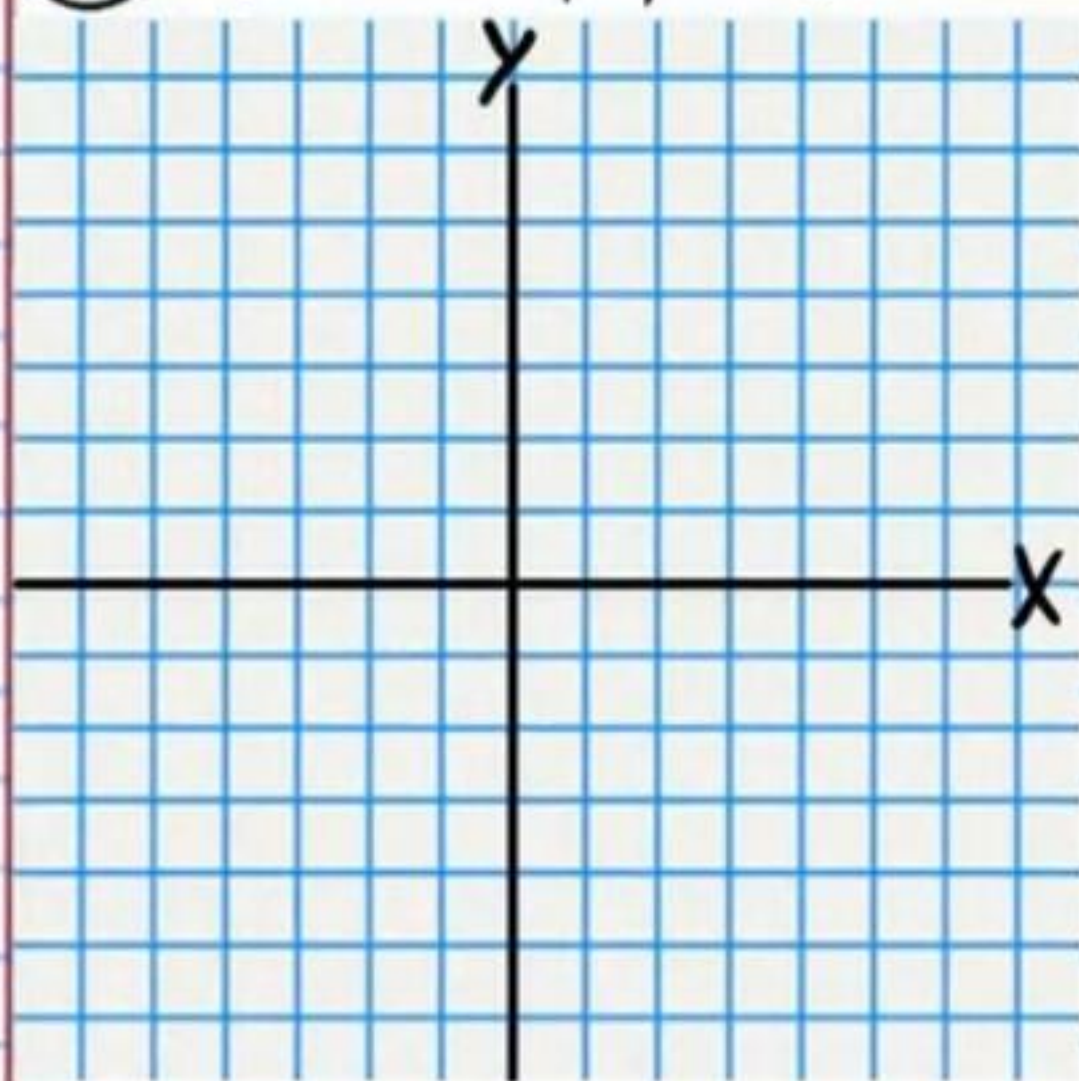
$$\frac{y}{x} - x = 5$$

Graph the following equations. Write the coordinates of at least 4 points. Use a graphing calculator to draw the other parts of the plot. Then eliminate the parameter.

②② $x = t + 1$, $y = 2t^3$

Point #1

Point #2



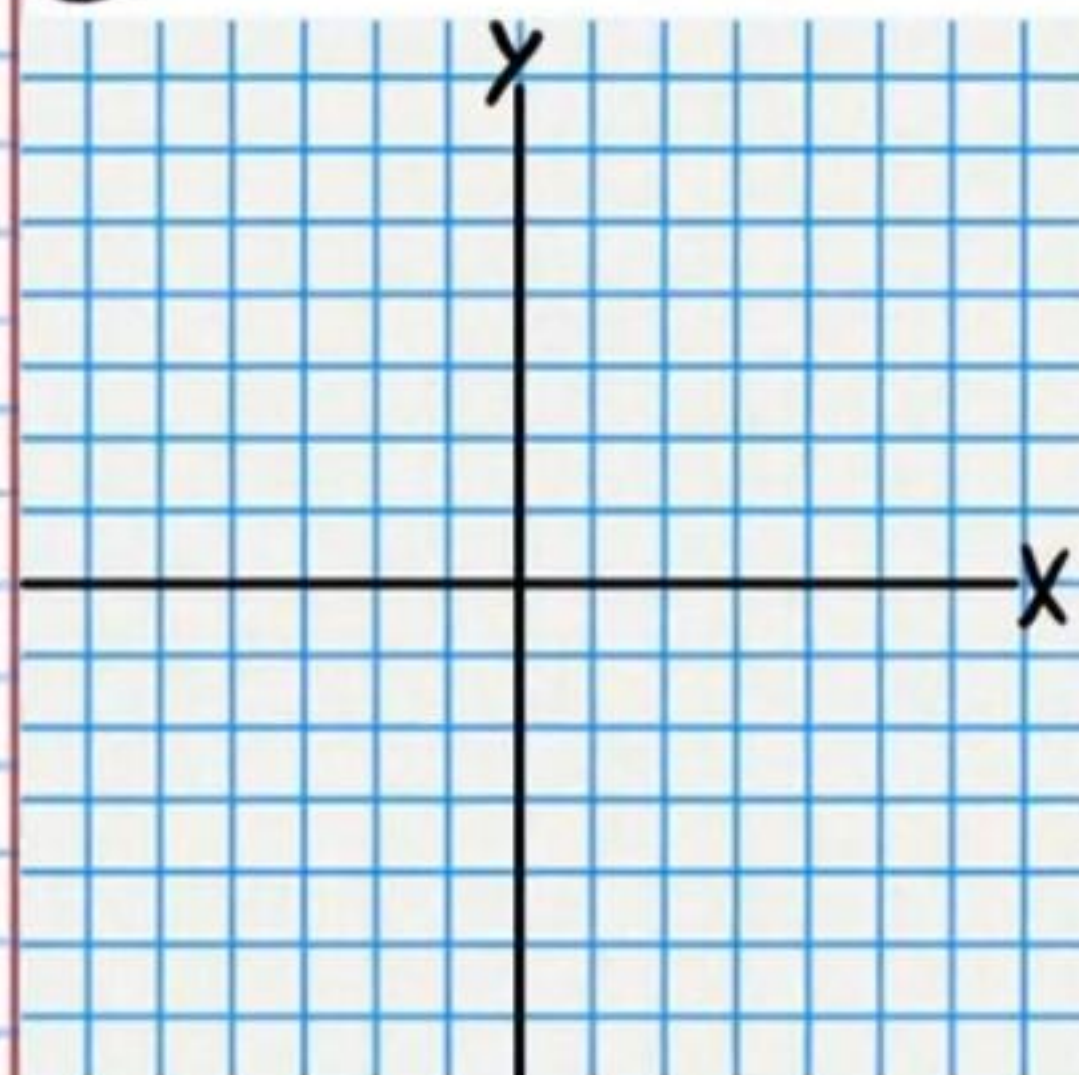
Point #3

Eliminate the parameter

$$x = t + 1, y = 2t^3$$

Point #4

②③ $x = 3 - 2\cos t$, $y = 1 + 5\sin t$ Point #1 Point #2



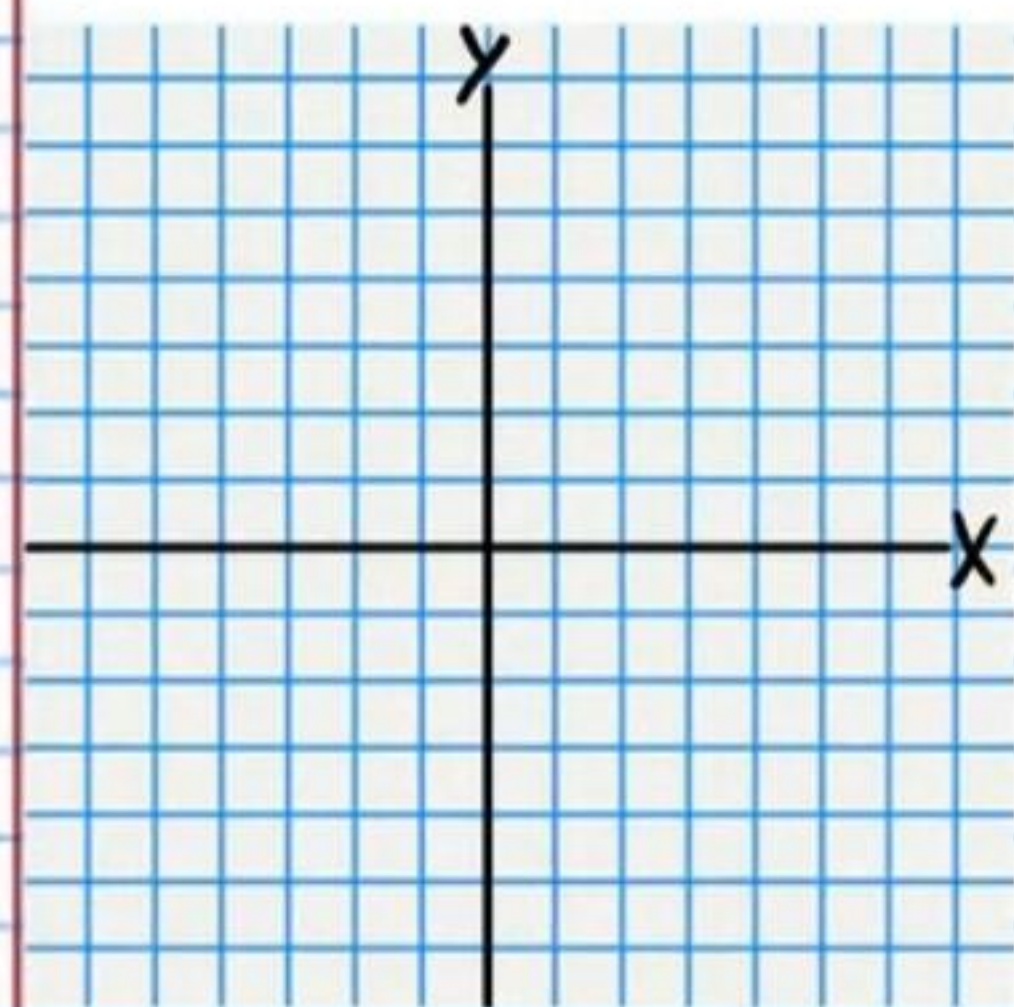
Point #3

Eliminate the parameter

$$x = 3 - 2\cos t, y = 1 + 5\sin t$$

Point #4

②④ $x = \cos^3 t$, $y = 1 - \cos t$ Point #1 Point #2



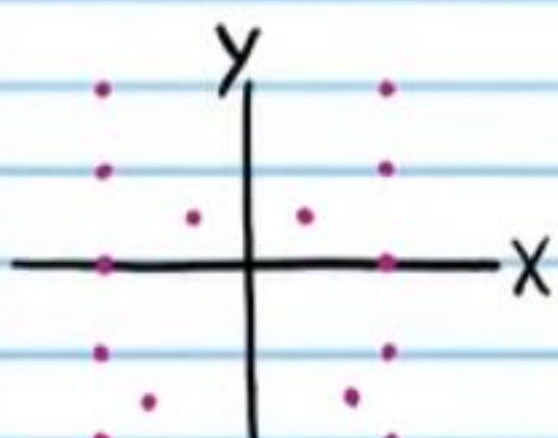
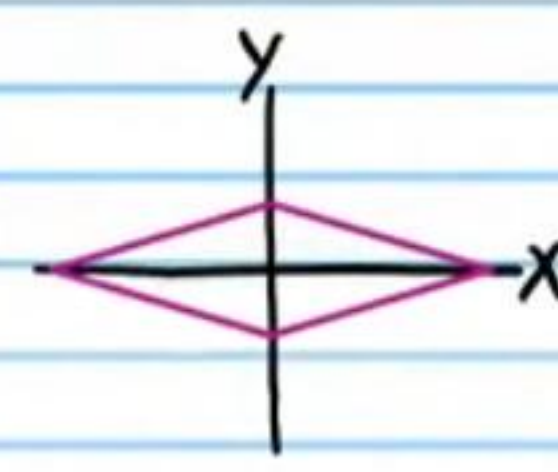
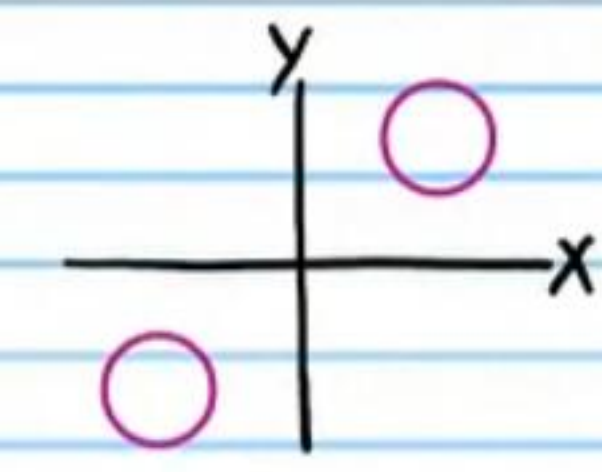
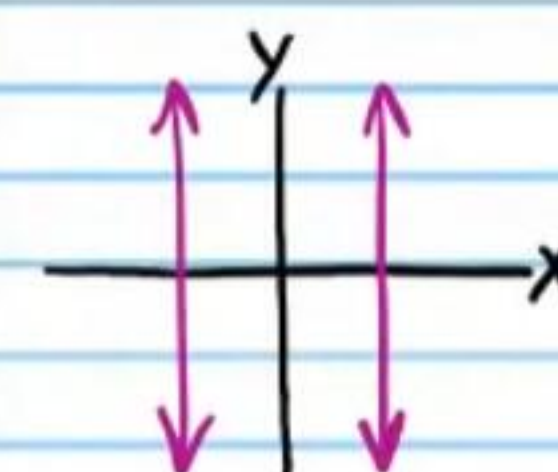
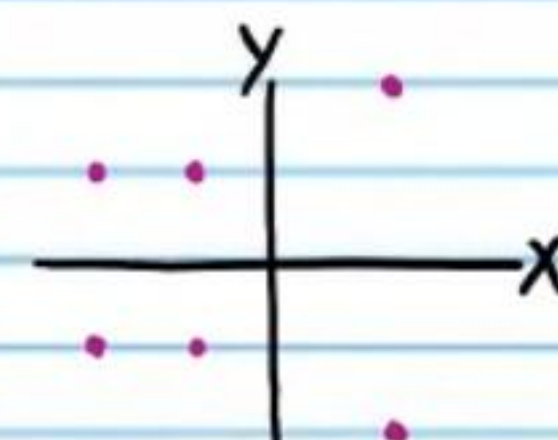
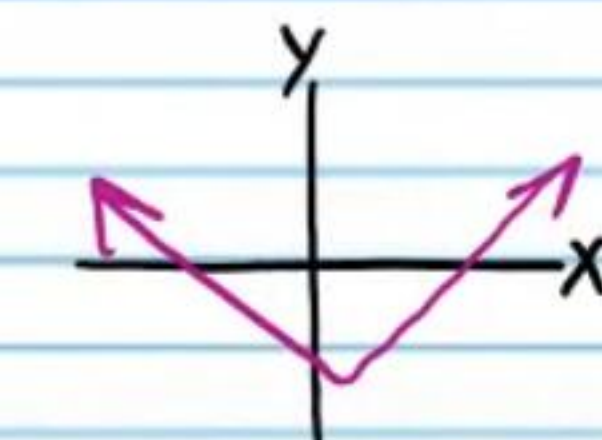
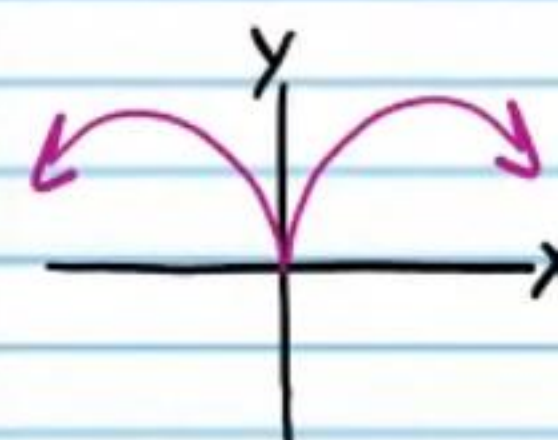
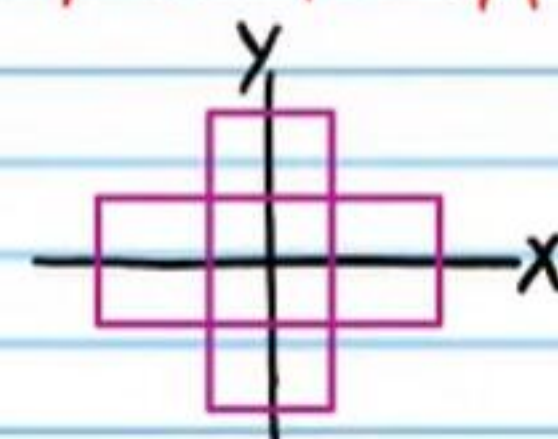
Point #3

Eliminate the parameter

$x = \cos^3 t$, $y = 1 - \cos t$

Point #4

②⑤ Relations are written below in the form of graphs. If a relation appears to be symmetric about the x-axis, write "x." If a relation appears to be symmetric about the y-axis, write "y." If a relation appears to be symmetric about the origin, write "O." If the relation shows no apparent symmetry, write "N." Some graphs might display multiple types of symmetry.



Test III Answers

① i)

X	2	-2	3	-3	4	2	-2	3	-3	4
Y	-2	2	-3	3	-4	2	-2	3	-3	4

iii) Yes

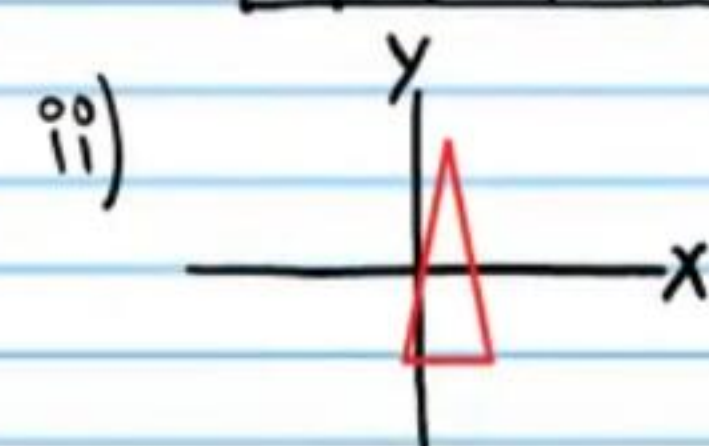
ii)

X	9	11	-2	-1	14	-9	-11	2	1	-14
Y	-9	-11	0	3	-14	-9	-11	0	3	-14

iv) Yes

⑤ i)

X	40	33	4	-11	21
Y	100	22	-7	-11	20



iii) $2^x = y + 18$

iv) $|y - 4| = |x - 1|$

② i) $\frac{y}{\sqrt{x^2 + y^2}} = \sqrt{x^2 + y^2}$

or $y = x^2 + y^2$

ii) $f(-x) = 3^{-x} + 4$

Neither

③ i) $x = t, y = 4^t + 1$

ii) $f(-x) = \frac{1}{(-x)^2} + 10(-x)^4$

$f(-x) = \frac{1}{x^2} + 10x^4$

Even

⑦

i)

X	6	10	-10	-16	-2	-6	-10	10	16	2
f(x)	-1	2	0	4	-3	-1	2	0	4	-3

X	15	9	-3	-8	-7	-15	-9	3	8	7
h(x)	14	5	-3	-8	-1	-14	-9	3	8	1

④ i) $\frac{\csc x}{x} + \frac{y}{\cot y} = 1$

ii) $\frac{\csc x}{x} + \frac{y}{\cot y} = 1$

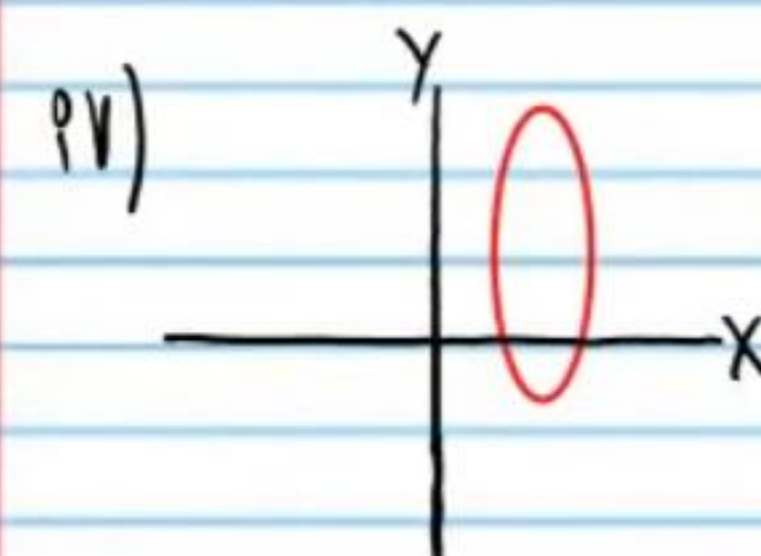
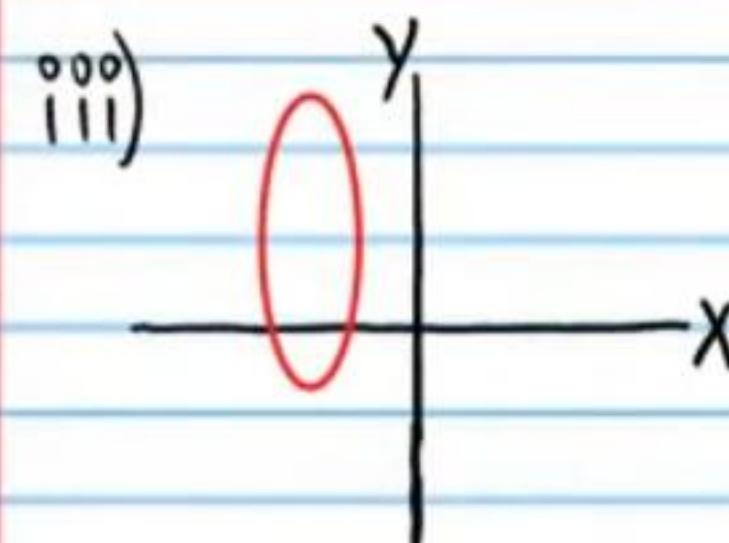
ii) Odd Odd Odd Neither
Even Neither Even Neither

⑥ i)

X	23	44	-6	-20	-3
Y	-34	-20	2	30	-19

ii)

X	-23	-44	6	20	3
Y	34	20	-2	-30	19



⑨ $x \approx 0.43$

⑩ $(8, -8)$

⑪ $(0, -5)$

⑫ $(\frac{7}{2}, \frac{7}{2}\sqrt{3})$ or $(3.5, 3.5\sqrt{3})$

⑬ $(9.205, 3.907)$

⑭ Other answers are possible.

$(-\sqrt{5}, 63.435^\circ)$ or

$(\sqrt{5}, 243.435^\circ)$ or

$(\sqrt{5}, -116.565^\circ)$

⑮ Other answers are possible.

$(\sqrt{13}, -33.69^\circ)$

or $(-\sqrt{13}, 146.31^\circ)$

or $(\sqrt{13}, 326.31^\circ)$

⑯ Other answers are possible.

$(18, 150^\circ)$

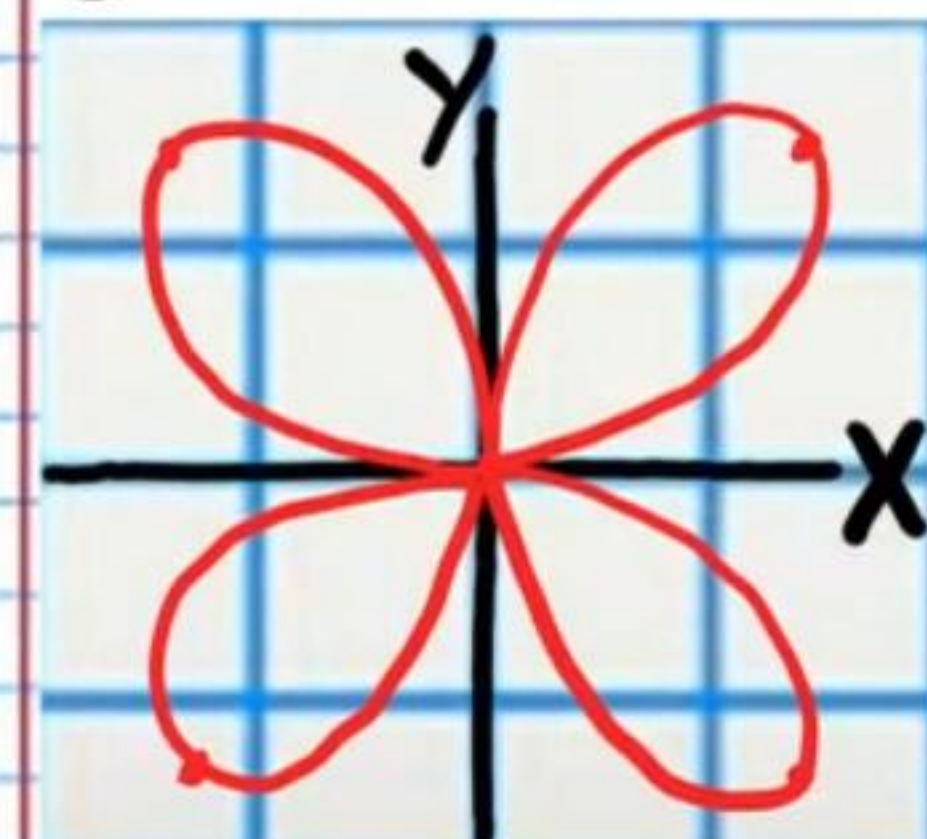
or $(-18, -30^\circ)$

or $(18, -210^\circ)$

⑰ Other answers are possible.

$(11, 0^\circ)$ or $(11, 360^\circ)$ or $(11, -360^\circ)$

18

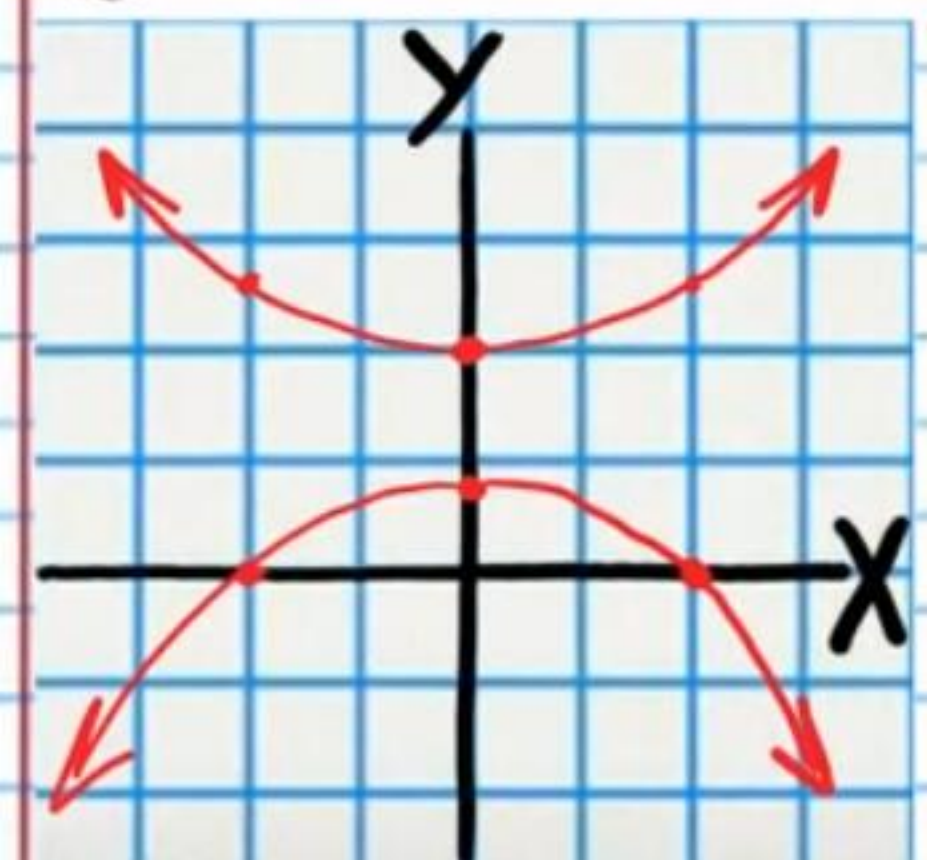


Other points are possible. Only 4 are necessary.

$$(2, 45^\circ) \quad (-2, -45^\circ) \quad (0, 0)$$

$$(-2, 135^\circ) \quad (2, -135^\circ)$$

19

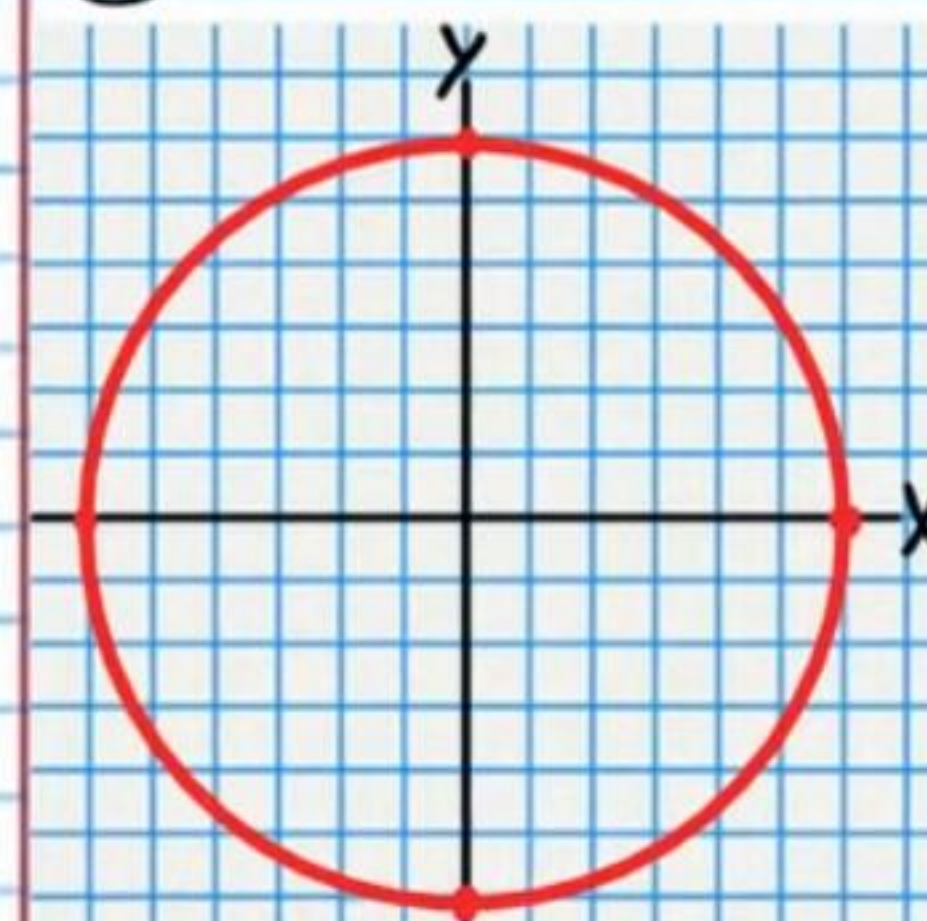


Other points are possible. Only 4 are necessary.

$$(2, 0^\circ) \quad \left(\frac{2}{3}, 90^\circ\right) \quad (-2, -90^\circ)$$

$$(2, 180^\circ) \quad (2, -180^\circ) \quad (1, 30^\circ)$$

20



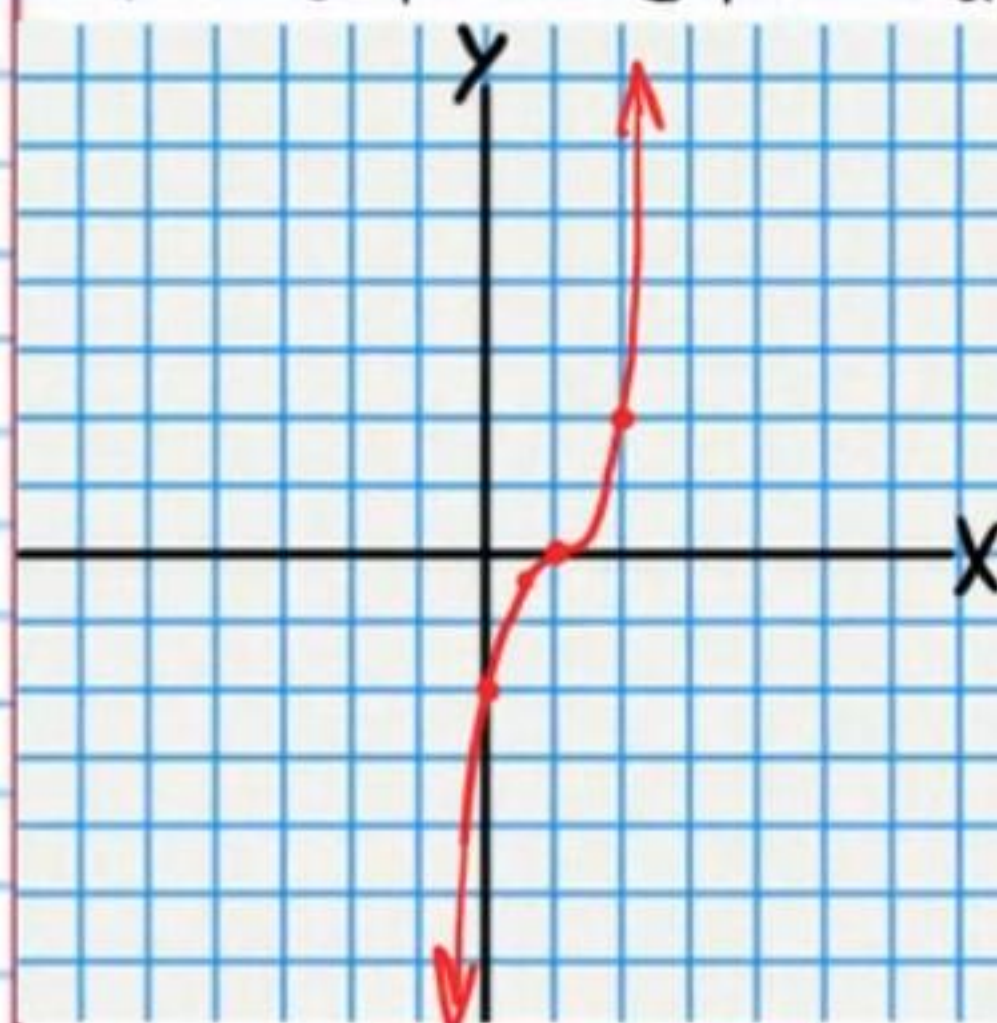
Other points are possible. Only 4 are necessary.

$$(6, 0) \quad (0, -6) \quad (0, 6) \quad (-6, 0)$$

21) $\tan \theta - r \cos \theta = 5$

22) Other points are possible. Only 4 are necessary.

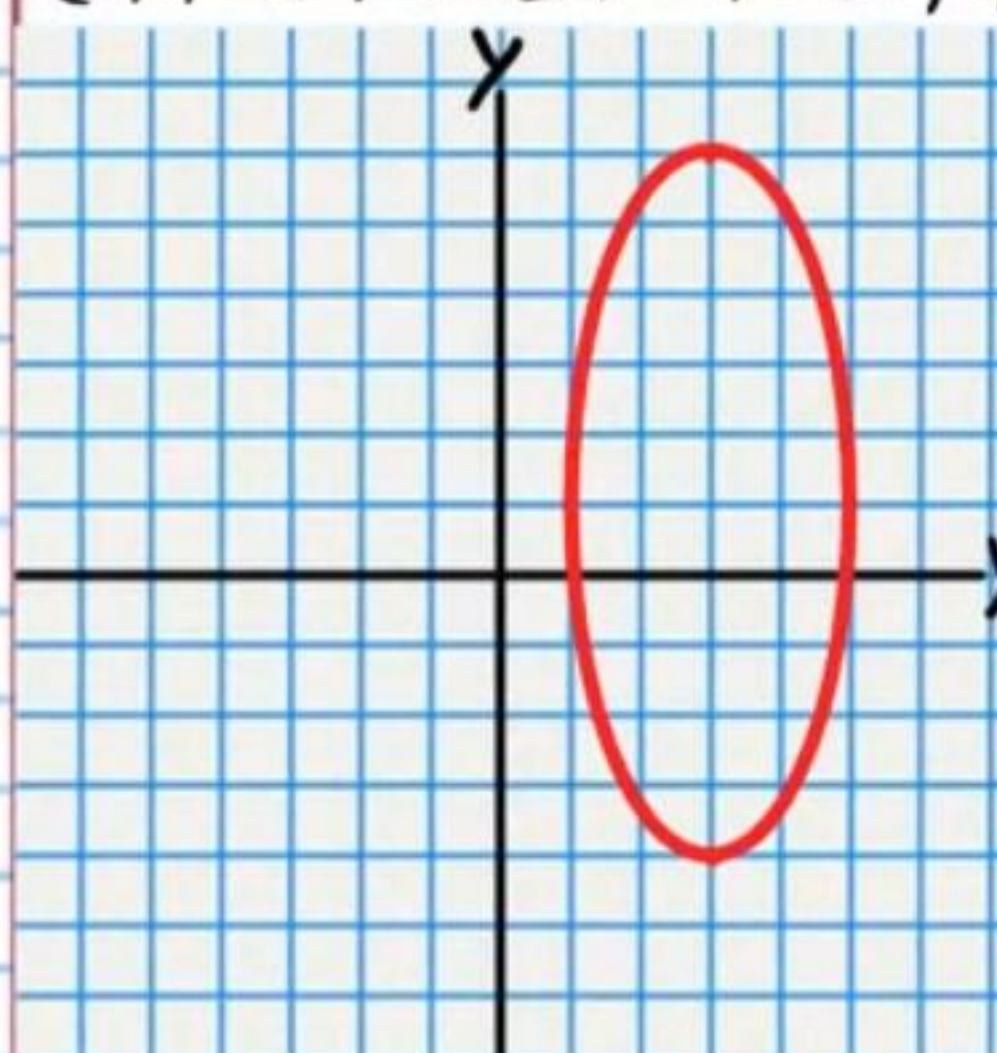
$$(1, 0) \quad (0, -2) \quad (2, 2) \quad \left(\frac{1}{2}, -\frac{1}{4}\right)$$



$$y = 2(x-1)^3$$

23) Other points are possible. Only 4 are necessary.

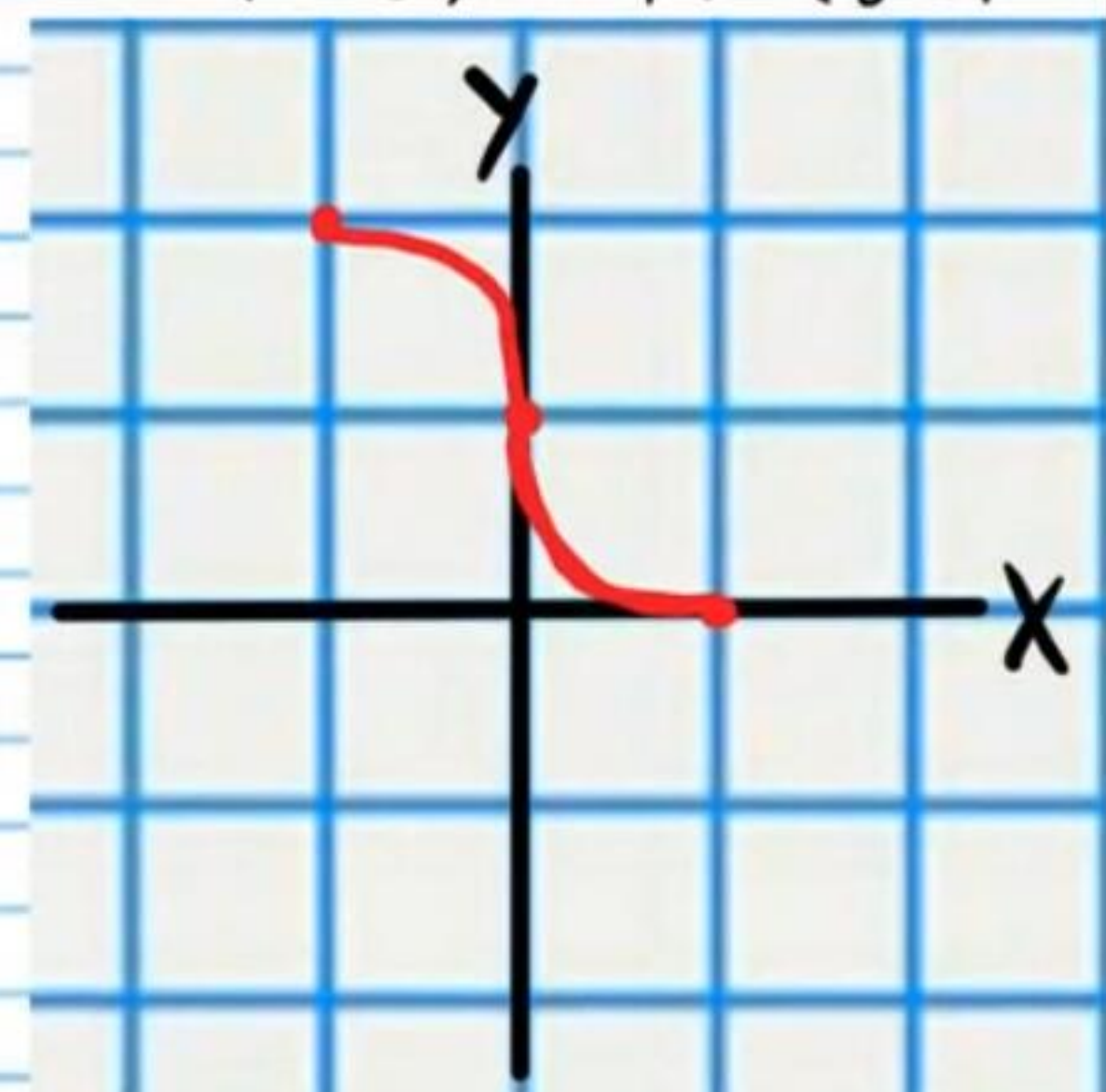
$$(1, 1) \quad (3, 6) \quad (3, -4) \quad (5, 1)$$



$$\frac{(x-3)^2}{4} + \frac{(y-1)^2}{25} = 1$$

24) Other points are possible. Only 4 are necessary.

$$(1, 0) \quad (0, 1) \quad (-1, 2) \quad \left(\frac{1}{8}, \frac{1}{2}\right)$$



$$x = (1-y)^3 \quad \text{or} \quad x = -(y-1)^3$$

25) N $x \& y \& 0$ N

y N x

$x \& y \& 0$ 0 $x \& y \& 0$

y N N